Harborlight Savings CECL Gap Assessment

March 13, 2026

Executive Summary

This assessment covers Harborlight Savings's Q1 2026 CECL Readiness Gap Assessment package. The materials are sufficient for a structured documentation-led gap assessment, but they do not support an execution-based CECL review because no reserve engine, reproducibility notebook, or lineaged execution runbook was supplied. The review therefore focuses on evidence sufficiency, scenario-definition consistency, provided-output reconciliation, and overlay documentation support.

Scope and Coverage Boundary

This work assessed whether the supplied CECL package is sufficient to support a deeper reserve review and whether the materials provided are internally consistent. The review did not attempt baseline reproduction, scenario reruns, or sensitivity testing because the implementation artifacts required for those procedures were not part of the package.

- Supported work: evidence sufficiency review, scenario-definition consistency review, provided-output reconciliation, and overlay documentation review.
- Blocked work: reserve-engine execution, baseline reproduction, implementation sensitivity testing, and model-code review.
- Coverage note: all conclusions are bounded to the documents, scenario tables, and output snapshots provided in the package.

Bank-Uploaded Input Package

- data/data_dictionary.csv (dataset)
- data/sample_rollforward.csv (dataset)
- docs/evidence_request_log.md (document)
- docs/gap_tracker.md (document)
- docs/governance_minutes.md (document)
- docs/methodology.md (document)

- docs/model_overview.md (document)
- docs/overlay_memo.md (document)
- docs/prior_review_note.md (document)
- docs/scenario_assumptions.md (document)
- outputs/provided_overlay_bridge.csv (packaged output table)
- outputs/provided_reserve_summary.csv (packaged output table)
- outputs/provided_segment_reserves.csv (packaged output table)
- scenarios/adverse.csv (scenario table)
- scenarios/baseline.csv (scenario table)
- scenarios/severe.csv (scenario table)

Artifact Provenance and Review Record

This report uses explicit provenance markers so a reviewer can distinguish what the bank supplied from what Codex produced during the review. [BANK INPUT] indicates a file contained in the uploaded package under input_package/. [CODEX OUTPUT] indicates a review artifact created during discovery, planning, gap assessment, or report rendering.

Bank-Uploaded Artifacts

- **E001 [BANK INPUT]** `data/data_dictionary.csv` – dataset; CSV artifact with 13 data row(s); Generated as part of the Codex review record for this case.
- **E002 [BANK INPUT]** `data/sample_rollforward.csv` – dataset; CSV artifact with 4 data row(s); Generated as part of the Codex review record for this case.
- **E003 [BANK INPUT]** `docs/evidence_request_log.md` – document; Text artifact with 6 line(s); Package inventory and coverage classification: Determine whether the upload supports execution-based review or only a documentation-led assessment.; Execution readiness and lineage assessment: Establish whether the package includes the artifacts required for baseline reproduction and scenario reruns.; Baseline reproduction: Reproduce the baseline reserve from the implementation and baseline scenario.; Evidence request and remediation planning: Translate identified blockers and inconsistencies into a prioritized request list for a future execution-based review.
- **E004 [BANK INPUT]** `docs/gap_tracker.md` – document; Text artifact with 7 line(s); Package inventory and coverage classification: Determine whether the upload supports execution-based review or only a documentation-led assessment.; Execution readiness and lineage assessment: Establish whether the package includes the artifacts required for baseline reproduction and scenario reruns.; Scenario reruns and sensitivity testing: Rerun scenarios and challenge reserve sensitivity to key assumptions.

- **E005 [BANK INPUT]** `docs/governance_minutes.md` – document; Text artifact with 3 line(s); Execution readiness and lineage assessment: Establish whether the package includes the artifacts required for baseline reproduction and scenario reruns.
- **E006 [BANK INPUT]** `docs/methodology.md` – document; Text artifact with 9 line(s); Horizon and reversion consistency review: Test whether documented forecast and reversion descriptions reconcile across methodology, model overview, and supplied scenario files.; Segment taxonomy reconciliation: Compare documented segment definitions to the segment structure in the provided reserve outputs.
- **E007 [BANK INPUT]** `docs/model_overview.md` – document; Text artifact with 3 line(s); Horizon and reversion consistency review: Test whether documented forecast and reversion descriptions reconcile across methodology, model overview, and supplied scenario files.; Segment taxonomy reconciliation: Compare documented segment definitions to the segment structure in the provided reserve outputs.
- **E008 [BANK INPUT]** `docs/overlay_memo.md` – document; Text artifact with 3 line(s); Overlay magnitude reconciliation: Compare documented overlay posture to the reserve bridge included in the package.
- **E009 [BANK INPUT]** `docs/prior_review_note.md` – document; Text artifact with 3 line(s); Generated as part of the Codex review record for this case.
- **E010 [BANK INPUT]** `docs/scenario_assumptions.md` – document; Text artifact with 3 line(s); Scenario narrative versus table comparison: Test whether the severe scenario narrative matches the supplied numeric scenario path.; Horizon and reversion consistency review: Test whether documented forecast and reversion descriptions reconcile across methodology, model overview, and supplied scenario files.
- **E011 [BANK INPUT]** `outputs/provided_overlay_bridge.csv` – packaged output table; CSV artifact with 4 data row(s); Overlay magnitude reconciliation: Compare documented overlay posture to the reserve bridge included in the package.
- **E012 [BANK INPUT]** `outputs/provided_reserve_summary.csv` – packaged output table; CSV artifact with 3 data row(s); Package inventory and coverage classification: Determine whether the upload supports execution-based review or only a documentation-led assessment.; Baseline reproduction: Reproduce the baseline reserve from the implementation and baseline scenario.
- **E013 [BANK INPUT]** `outputs/provided_segment_reserves.csv` – packaged output table; CSV artifact with 12 data row(s); Segment taxonomy reconciliation: Compare documented segment definitions to the segment structure in the provided reserve outputs.
- **E014 [BANK INPUT]** `scenarios/adverse.csv` – scenario table; CSV artifact with 6 data row(s); Scenario narrative versus table comparison: Test whether the severe scenario narrative matches the supplied numeric scenario path.; Scenario reruns and sensitivity testing: Rerun scenarios and challenge reserve sensitivity to key assumptions.
- **E015 [BANK INPUT]** `scenarios/baseline.csv` – scenario table; CSV artifact with 6 data row(s); Scenario reruns and sensitivity testing: Rerun scenarios and challenge reserve sensitivity to key assumptions.

- **E016 [BANK INPUT]** `scenarios/severe.csv` – scenario table; CSV artifact with 6 data row(s); Scenario narrative versus table comparison: Test whether the severe scenario narrative matches the supplied numeric scenario path.; Scenario reruns and sensitivity testing: Rerun scenarios and challenge reserve sensitivity to key assumptions.

Codex-Generated Review Artifacts

- **E017 [CODEX OUTPUT]** `outputs/support/discovery_summary.json` – json artifact; Text artifact with 80 line(s); Generated as part of the Codex review record for this case.
- **E018 [CODEX OUTPUT]** `outputs/support/discovery_summary.md` – document; Text artifact with 31 line(s); Summarizes what Codex identified in the uploaded package during discovery.
- **E019 [CODEX OUTPUT]** `outputs/support/case_understanding.md` – document; Text artifact with 27 line(s); Records Codex's synthesized view of the CECL process, assumptions, and constraints.
- **E020 [CODEX OUTPUT]** `outputs/support/review_plan.md` – document; Text artifact with 21 line(s); Captures the planned review procedures before detailed execution.
- **E021 [CODEX OUTPUT]** `outputs/support/review_strategy.md` – document; Text artifact with 75 line(s); Explains why specific procedures were selected from the discovered evidence.
- **E022 [CODEX OUTPUT]** `outputs/support/executed_test_matrix.md` – document; Text artifact with 13 line(s); Registers which procedures were executed, blocked, and what each procedure concluded.
- **E023 [CODEX OUTPUT]** `outputs/support/executed_test_matrix.csv` – generated review table; CSV artifact with 9 data row(s); Machine-readable form of the executed procedure register.
- **E024 [CODEX OUTPUT]** `outputs/support/evidence_map.md` – document; Text artifact with 78 line(s); Maps uploaded and derived artifacts to the procedures they supported.
- **E025 [CODEX OUTPUT]** `outputs/support/agentic_review_log.md` – document; Text artifact with 43 line(s); Readable stage-level summary of the intended Codex review flow.
- **E026 [CODEX OUTPUT]** `outputs/support/procedure_run_log.md` – document; Text artifact with 83 line(s); Chronological log of granular review actions and outputs.
- **E027 [CODEX OUTPUT]** `outputs/support/procedure_run_log.csv` – generated review table; CSV artifact with 10 data row(s); Machine-readable chronology of granular review actions.
- **E028 [CODEX OUTPUT]** `outputs/support/evidence_excerpts.md` – document; Text artifact with 133 line(s); Collects raw excerpts from uploaded materials and review outputs used in the writeup.
- **E029 [CODEX OUTPUT]** `outputs/support/documentation_crosscheck.md` – document; Text artifact with 23 line(s); Stores documentation-versus-implementation and output cross-check observations.
- **E030 [CODEX OUTPUT]** `outputs/support/findings_register.json` – json artifact; Text artifact with 43 line(s); Registers findings, severity, and supporting evidence references.

- **E031** [**CODEX OUTPUT**] `outputs/support/analysis_summary.json` – json artifact; Text artifact with 5 line(s); Generated as part of the Codex review record for this case.
- **E032** [**CODEX OUTPUT**] `outputs/support/evidence_ledger.json` – json artifact; Text artifact with 346 line(s); Generated as part of the Codex review record for this case.
- **E033** [**CODEX OUTPUT**] `outputs/support/coverage_statement.md` – document; Text artifact with 15 line(s); States supported versus blocked review coverage.
- **E034** [**CODEX OUTPUT**] `outputs/support/evidence_request_list.md` – document; Text artifact with 6 line(s); Generated as part of the Codex review record for this case.
- **E035** [**CODEX OUTPUT**] `outputs/support/provided_reserve_summary.csv` – generated review table; CSV artifact with 3 data row(s); Generated as part of the Codex review record for this case.
- **E036** [**CODEX OUTPUT**] `outputs/support/provided_segment_reserves.csv` – generated review table; CSV artifact with 12 data row(s); Generated as part of the Codex review record for this case.
- **E037** [**CODEX OUTPUT**] `outputs/support/provided_overlay_bridge.csv` – generated review table; CSV artifact with 4 data row(s); Generated as part of the Codex review record for this case.
- **E038** [**CODEX OUTPUT**] `outputs/support/codex_trace.md` – document; Text artifact with 27 line(s); Readable execution trace of discovery, planning, execution, and synthesis.

Uploaded Package Tree

```
input_package/
  data/
    data_dictionary.csv
    sample_rollforward.csv
  docs/
    evidence_request_log.md
    gap_tracker.md
    governance_minutes.md
    methodology.md
    model_overview.md
    overlay_memo.md
    prior_review_note.md
    scenario_assumptions.md
  outputs/
    provided_overlay_bridge.csv
    provided_reserve_summary.csv
    provided_segment_reserves.csv
  scenarios/
    adverse.csv
    baseline.csv
    severe.csv
```

Codex Output Tree

```
outputs/
  support/
    agentic_review_log.md
    analysis_summary.json
    case_understanding.md
    codex_trace.md
    coverage_statement.md
    discovery_summary.json
    discovery_summary.md
    documentation_crosscheck.md
    evidence_excerpts.md
    evidence_ledger.json
    evidence_map.md
    evidence_request_list.md
    executed_test_matrix.csv
    executed_test_matrix.md
    findings_register.json
    procedure_run_log.csv
    procedure_run_log.md
    provided_overlay_bridge.csv
    provided_reserve_summary.csv
    provided_segment_reserves.csv
    review_plan.md
    review_strategy.md
```

Package Discovery and Case Understanding

The discovered package is best understood as a documentation-led CECL gap assessment for Harborlight Savings's Q1 2026 CECL Readiness Gap Assessment. It contains narrative methodology, scenario definitions, prior output snapshots, and governance artifacts, but it does not contain the implementation artifacts required for execution-based validation.

- Central assumption or theme: Methodology documents a 8-quarter forecast and 4-quarter reversion.
- Central assumption or theme: Scenario narratives should align quarter by quarter to numeric scenario files and to any reserve conclusions drawn from them.
- Central assumption or theme: Documented segment taxonomy and overlay limits should reconcile to the provided reserve outputs.
- Reviewable scope: Documentation sufficiency and internal consistency review.
- Reviewable scope: Scenario narrative versus numeric table comparison.
- Reviewable scope: Provided-output segment and overlay reconciliation.
- Reviewable scope: Evidence request formulation for a future execution-based review.
- Key risk to challenge: Scenario documentation may overstate severity relative to the numeric scenario path.

- Key risk to challenge: Segment taxonomy may not reconcile across documents and outputs.
- Key risk to challenge: Overlay governance may be understated relative to the provided reserve bridge.
- Constraint: No reserve engine was supplied.
- Constraint: No reproducibility notebook or execution runbook was supplied.
- Constraint: No lineaged link from outputs to a controlled model version was supplied.

Documented CECL Process Understanding

The package describes a CECL process covering 5 documented segments: Residential Mortgage, Heloc, Cre Investor, Cre Owner Occupied, Commercial And Industrial. Documentation frames the workflow as a narrative and prior-output review package rather than a reproducible model handoff.

Assumption Area	Documented Position	Observed in Package	Review Implication
Forecast horizon	8 quarters in methodology	6 quarters in model overview / 6 numeric quarters provided	Horizon treatment is not consistently documented or evidenced.
Reversion horizon	4 quarters in methodology	2 quarters in model overview / no reversion files provided	Reversion mechanics cannot be independently assessed.
Segment taxonomy	5 documented segments	4 output segments in reserve files	Output reporting does not reconcile to the documented segment structure.
Overlay guardrail	6.0 bps documented cap	18.0 bps shown in output bridge	Qualitative adjustment support is incomplete and inconsistent.

Review Strategy and Planning Logic

The review strategy was shaped by the discovered package boundary. Codex considered 4 central review questions, executed 7 documentation-led procedures, and marked 2 procedures as blocked because the package lacked runnable implementation artifacts. Because the package lacks a reserve engine and execution lineage, the review strategy intentionally shifts from runtime validation to evidence sufficiency, scenario consistency, output reconciliation, and remediation planning. The goal is to show what can be credibly reviewed now and what remains blocked.

1. Does the package support a model-driven CECL opinion, or only a documentation-led gap assessment?
2. Are the scenario narratives and numeric scenario files aligned closely enough to support directional reserve conclusions?
3. Do documented forecast horizon, reversion, segment taxonomy, and overlay guardrails reconcile to the provided output snapshots?
4. What concrete evidence must be supplied before baseline reproduction, scenario reruns, and sensitivity testing can be performed credibly?

ID	Procedure	Status	Why Selected
GA-01	Package inventory and coverage classification	Executed	The package contained scenario files, output snapshots, and governance documents but no reserve engine or runtime lineage.
GA-02	Execution readiness and lineage assessment	Executed	A clear coverage boundary is necessary before any deeper opinion can be formed.
GA-03	Scenario narrative versus table comparison	Executed	Scenario definition quality is reviewable even without runnable code.
GA-04	Horizon and reversion consistency review	Executed	Horizon and reversion are governance-critical CECL assumptions even when the implementation is absent.
GA-05	Segment taxonomy reconciliation	Executed	Output reconciliation is one of the few quantitative checks available without the model.
GA-06	Overlay magnitude reconciliation	Executed	Overlay support is reviewable from documentation and provided output snapshots even without code.
GA-07	Baseline reproduction	Blocked	This would normally be a core CECL procedure if the package contained runnable implementation artifacts.
GA-08	Scenario reruns and sensitivity testing	Blocked	These procedures were considered because they would normally follow scenario and methodology review.
GA-09	Evidence request and remediation planning	Executed	The package still supports useful next-step guidance even without runnable code.

Selected Raw Evidence Excerpts

The excerpts below are short verbatim snippets from bank-uploaded materials and Codex-generated outputs that support the current gap assessment and coverage boundary.

[BANK INPUT] docs/methodology.md

Documented forecast, reversion, segmentation, and missing-runtime boundary used in the gap assessment.

```
# CECL Methodology

The documented CECL process for Harborlight Savings describes a 8-quarter reasonable-and-supportable pe

- Residential Mortgage
- HELOC
- CRE Investor
- CRE Owner Occupied
- Commercial and Industrial
```

[BANK INPUT] docs/scenario_assumptions.md

Numeric scenario horizon and narrative-alignment discussion showing the mismatch between documented and provided scenario depth.

```
# Scenario Assumptions

Severe is described as a uniformly harsher path than adverse across the full review horizon. Management
```

[BANK INPUT] docs/overlay_memo.md

Overlay cap language and segment overlay values that Codex challenged against the provided reserve bridge.

```
# Overlay Memo

Qualitative overlay is described as modest and capped at 6.0 bps, pending additional governance evidenc
```

[BANK INPUT] outputs/provided_overlay_bridge.csv + outputs/provided_segment_reser

Bank-supplied output snapshots used for the documentation-led reconciliation work.

```
[provided_overlay_bridge.csv]
segment_id,documented_cap_bps,provided_overlay_bps
residential_mortgage,6.0,12.0
heloc,6.0,16.0
cre_investor,6.0,18.0
commercial_and_industrial,6.0,14.0

[provided_segment_reserves.csv]
segment_id,scenario_name,reserve_amount
residential_mortgage,baseline,8420000.0
residential_mortgage,adverse,11300000.0
residential_mortgage,severe,10980000.0
heloc,baseline,5180000.0
heloc,adverse,6450000.0
heloc,severe,6430000.0
cre_investor,baseline,11900000.0
cre_investor,adverse,17120000.0
cre_investor,severe,17880000.0
commercial_and_industrial,baseline,12920000.0
commercial_and_industrial,adverse,17310000.0
commercial_and_industrial,severe,17720000.0
```

[CODEX OUTPUT] outputs/support/coverage_statement.md

Codex-generated coverage boundary showing which procedures were supported and which were blocked.

```
# Coverage Statement

The package contains useful CECL documentation and output snapshots, but it does not contain the impleme

## Supported Work
```

```
- Discovery and evidence sufficiency review
- Scenario-definition consistency review
- Provided-output reconciliation
- Overlay documentation review

## Blocked Work
- Baseline reproduction
- Scenario reruns against a reserve engine
- Sensitivity testing on implementation assumptions
- Model-code review
```

[CODEX OUTPUT] outputs/support/findings_register.json

Codex-generated findings register capturing the primary documentation and evidence gaps.


```
{
  "findings": [
    {
      "severity": "high",
      "title": "Execution-based review is blocked by missing reserve engine and lineage evidence",
      "summary": "The package includes prior outputs and narrative materials, but no reserve engine, rep
      "evidence": [
        "docs/evidence_request_log.md",
        "docs/gap_tracker.md",
        "outputs/provided_reserve_summary.csv"
      ]
    },
    {
      "severity": "high",
      "title": "Scenario narrative is not fully aligned to the supplied scenario tables",
      "summary": "The severe scenario narrative describes a uniformly harsher path, but the supplied nun
      "evidence": [
        "docs/scenario_assumptions.md",
        "scenarios/adverse.csv",
        "scenarios/severe.csv"
      ]
    },
    {
      "severity": "medium",
      "title": "Documented segment structure does not reconcile to the provided reserve outputs",
      "summary": "Documentation describes 5 segments, while the provided reserve outputs reconcile to 4
      "evidence": [
        "docs/methodology.md",
        "docs/model_overview.md",
        "outputs/provided_segment_reserves.csv"
      ]
    },
    {
      "severity": "medium",
      "title": "Overlay documentation understates the magnitude implied by the provided reserve bridge"
      "summary": "The overlay memo frames qualitative adjustment as capped at 6.0 bps, but the provided
      "evidence": [
        "docs/overlay_memo.md",
```

```
        "outputs/provided_overlay_bridge.csv"
      ]
    }
  ]
}
```


Procedures Performed

GA-01 - Package inventory and coverage classification

Determine whether the upload supports execution-based review or only a documentation-led assessment.

- **Status:** executed
- **Why selected:** The package contained scenario files, output snapshots, and governance documents but no reserve engine or runtime lineage.
- **Bank-input evidence relied upon:** `docs/evidence_request_log.md`, `docs/gap_tracker.md`, `outputs/provided_reserve_summary.csv`
- **Codex-generated evidence relied upon:** *None*
- **Codex-generated outputs produced:** `discovery_summary.md`, `coverage_statement.md`
- **Assessment criteria:** Execution-based procedures should be blocked unless code, runbook, and reproducibility evidence are present.
- **Key result:** Case classified as documentation-led CECL gap assessment; execution-based review blocked.

GA-02 - Execution readiness and lineage assessment

Establish whether the package includes the artifacts required for baseline reproduction and scenario reruns.

- **Status:** executed
- **Why selected:** A clear coverage boundary is necessary before any deeper opinion can be formed.
- **Bank-input evidence relied upon:** `docs/evidence_request_log.md`, `docs/gap_tracker.md`, `docs/governance_minutes.md`
- **Codex-generated evidence relied upon:** *None*
- **Codex-generated outputs produced:** `coverage_statement.md`, `documentation_crosscheck.md`
- **Assessment criteria:** Reserve engine, reproducibility materials, and run lineage must be available before execution work can proceed.
- **Key result:** Runtime validation remained blocked because no reserve engine, reproducibility notebook, or lineaged runbook was supplied.

GA-03 - Scenario narrative versus table comparison

Test whether the severe scenario narrative matches the supplied numeric scenario path.

- **Status:** executed
- **Why selected:** Scenario definition quality is reviewable even without runnable code.
- **Bank-input evidence relied upon:** `docs/scenario_assumptions.md`, `scenarios/adverse.csv`, `scenarios/severe.csv`
- **Codex-generated evidence relied upon:** *None*
- **Codex-generated outputs produced:** `documentation_crosscheck.md`, `findings_register.json`
- **Assessment criteria:** A narrative described as uniformly harsher should be reflected in the quarter-by-quarter numeric path or explicitly qualified.
- **Key result:** Severe house-price growth was less severe than adverse in 4 quarters: 2026Q3, 2026Q4, 2027Q1, 2027Q2.

GA-04 - Horizon and reversion consistency review

Test whether documented forecast and reversion descriptions reconcile across methodology, model overview, and supplied scenario files.

- **Status:** executed
- **Why selected:** Horizon and reversion are governance-critical CECL assumptions even when the implementation is absent.
- **Bank-input evidence relied upon:** `docs/methodology.md`, `docs/model_overview.md`, `docs/scenario_ass`
- **Codex-generated evidence relied upon:** *None*
- **Codex-generated outputs produced:** `documentation_crosscheck.md`, `findings_register.json`
- **Assessment criteria:** Documents should describe a consistent forecast and reversion framework that is evidenced in supplied files.
- **Key result:** Methodology, model overview, and scenario files do not describe one consistent horizon/reversion structure.

GA-05 - Segment taxonomy reconciliation

Compare documented segment definitions to the segment structure in the provided reserve outputs.

- **Status:** executed
- **Why selected:** Output reconciliation is one of the few quantitative checks available without the model.
- **Bank-input evidence relied upon:** `docs/methodology.md`, `docs/model_overview.md`, `outputs/provided_`

- **Codex-generated evidence relied upon:** *None*
- **Codex-generated outputs produced:** `provided_segment_reserves.csv`, `findings_register.json`
- **Assessment criteria:** Documented segment structure should match the segmentation of provided output files or be explicitly bridged.
- **Key result:** Documentation references 5 segments, while outputs reconcile to 4 segments.

GA-06 - Overlay magnitude reconciliation

Compare documented overlay posture to the reserve bridge included in the package.

- **Status:** executed
- **Why selected:** Overlay support is reviewable from documentation and provided output snapshots even without code.
- **Bank-input evidence relied upon:** `docs/overlay_memo.md`, `outputs/provided_overlay_bridge.csv`
- **Codex-generated evidence relied upon:** *None*
- **Codex-generated outputs produced:** `provided_overlay_bridge.csv`, `findings_register.json`
- **Assessment criteria:** Documented overlay cap and rationale should reconcile to the magnitude shown in the reserve bridge.
- **Key result:** Overlay memo cites a 6.0 bps cap while the provided bridge reaches 18.0 bps.

GA-07 - Baseline reproduction

Reproduce the baseline reserve from the implementation and baseline scenario.

- **Status:** blocked
- **Why selected:** This would normally be a core CECL procedure if the package contained runnable implementation artifacts.
- **Bank-input evidence relied upon:** `outputs/provided_reserve_summary.csv`, `docs/evidence_request_l`
- **Codex-generated evidence relied upon:** *None*
- **Codex-generated outputs produced:** `coverage_statement.md`, `evidence_request_list.md`
- **Assessment criteria:** Cannot proceed without reserve engine, reproducibility materials, and execution lineage.
- **Key result:** Not executed because the package contains no reserve engine or reproducibility notebook.
- **Blocking factor:** Missing reserve engine and execution lineage.

GA-08 - Scenario reruns and sensitivity testing

Rerun scenarios and challenge reserve sensitivity to key assumptions.

- **Status:** blocked
- **Why selected:** These procedures were considered because they would normally follow scenario and methodology review.
- **Bank-input evidence relied upon:** `scenarios/baseline.csv`, `scenarios/adverse.csv`, `scenarios/severe.csv`, `docs/gap_tracker.md`
- **Codex-generated evidence relied upon:** *None*
- **Codex-generated outputs produced:** `coverage_statement.md`, `evidence_request_list.md`
- **Assessment criteria:** Cannot proceed without executable reserve logic, controllable parameters, and runbook support.
- **Key result:** Not executed because no model implementation or controllable parameter set was supplied.
- **Blocking factor:** Missing runtime artifacts and parameter lineage.

GA-09 - Evidence request and remediation planning

Translate identified blockers and inconsistencies into a prioritized request list for a future execution-based review.

- **Status:** executed
- **Why selected:** The package still supports useful next-step guidance even without runnable code.
- **Bank-input evidence relied upon:** `findings_register.json`, `coverage_statement.md`, `docs/evidence_request_log.md`
- **Codex-generated evidence relied upon:** *None*
- **Codex-generated outputs produced:** `evidence_request_list.md`
- **Assessment criteria:** Evidence requests should map directly to blocked procedures and identified gaps.
- **Key result:** Produced a focused evidence request list to unlock baseline reproduction, scenario reruns, and deeper model challenge.

Agentic Execution Workflow

Discovery

The review started by inventorying the package and determining whether it supported model execution or only documentation-led review.

- Review question addressed: What evidence is present, and what kind of CECL review does that evidence support?
- Decision or rationale: Treat the case as a gap assessment because the package contained narratives, scenarios, and outputs but no runnable reserve implementation.
- Inputs reviewed: input_package/*
- Outputs produced: discovery_summary.md, evidence_ledger.json, case_understanding.md

Case understanding and planning

Codex formed a narrowed review strategy based on the evidence boundary, explicitly separating executable work from blocked work.

- Review question addressed: Which review questions can be answered credibly now, and which must be deferred?
- Decision or rationale: Focus on evidence sufficiency, scenario consistency, segment reconciliation, and overlay support; mark runtime procedures as blocked.
- Inputs reviewed: discovery_summary.md, docs/evidence_request_log.md, docs/methodology.md, outputs/provided_reserve_summary.csv
- Outputs produced: review_plan.md, review_strategy.md, executed_test_matrix.md

Documentation and output cross-checking

The review challenged the package using the scenario tables, prior outputs, and governance materials that were actually supplied.

- Review question addressed: Do the narrative materials reconcile to the numeric scenario files and provided output snapshots?
- Decision or rationale: Escalate inconsistencies where the severe narrative does not match the numeric path, where segment definitions do not reconcile, or where overlay posture is understated.
- Inputs reviewed: docs/*.md, scenarios/*.csv, outputs/*.csv
- Outputs produced: documentation_crosscheck.md, findings_register.json, evidence_map.md

Blocked quantitative procedures

Codex explicitly considered baseline reproduction and sensitivity testing and recorded why they could not be executed.

- Review question addressed: What normally expected CECL validation work remains unsupported by the package?
- Decision or rationale: Do not simulate baseline reruns or sensitivities without a reserve engine, reproducibility notebook, or run lineage.

- Inputs reviewed: coverage_statement.md, executed_test_matrix.md, docs/gap_tracker.md
- Outputs produced: coverage_statement.md, evidence_request_list.md

Synthesis

The final stage combined discovery, planning, blocked-work logic, and documentation findings into a comprehensive internal gap assessment.

- Review question addressed: What opinion is actually supportable now, and what must management supply next?
- Decision or rationale: Issue a documentation-led gap assessment with explicit blockers, findings, and next-step evidence requests rather than overclaiming execution coverage.
- Inputs reviewed: findings_register.json, coverage_statement.md, evidence_request_list.md, agentic_review_log.md
- Outputs produced: cecl_gap_assessment.tex, cecl_gap_assessment.pdf

Chronological Procedure and Output Log

This section records the more granular action sequence for the gap assessment, including both executed documentation-led checks and explicitly blocked runtime procedures.

EVT-01 - Inventory uploaded package

- **Phase:** discovery
- **Related procedure:** GA-01
- **Action:** Enumerated the documentation package and checked whether a reserve engine, reproducibility notebook, and execution lineage were present.
- **Inputs:** `input_package/docs/*.md`, `input_package/scenarios/*.csv`, `input_package/outputs/*.csv`
- **Outputs:** `outputs/support/discovery_summary.md`, `outputs/support/evidence_ledger.json`
- **Result:** Case classified as documentation-led CECL gap assessment because runtime artifacts were not supplied.

EVT-02 - Define supported and blocked review lanes

- **Phase:** planning
- **Related procedure:** GA-01
- **Action:** Separated executable documentation-led checks from blocked runtime procedures before deeper review work.

- **Inputs:** `outputs/support/discovery_summary.md`, `input_package/docs/evidence_request_log.md`, `input_package/docs/gap_tracker.md`
- **Outputs:** `outputs/support/review_plan.md`, `outputs/support/review_strategy.md`, `outputs/support/c`
- **Result:** Evidence sufficiency, scenario consistency, output reconciliation, and overlay review retained; baseline reproduction and sensitivities blocked.

EVT-03 - Assess execution readiness and lineage

- **Phase:** documentation_challenge
- **Related procedure:** GA-02
- **Action:** Reviewed governance and evidence-request materials to determine whether the package could support model execution or reproduction.
- **Inputs:** `input_package/docs/evidence_request_log.md`, `input_package/docs/gap_tracker.md`, `input_package/docs/governance_minutes.md`
- **Outputs:** `outputs/support/coverage_statement.md`, `outputs/support/documentation_crosscheck.md`
- **Result:** Confirmed that no reserve engine, reproducibility notebook, or lineaged runbook was available.

EVT-04 - Compare scenario narrative to numeric scenario files

- **Phase:** documentation_challenge
- **Related procedure:** GA-03
- **Action:** Compared the severe scenario narrative to quarter-by-quarter severe and adverse scenario tables.
- **Inputs:** `input_package/docs/scenario_assumptions.md`, `input_package/scenarios/adverse.csv`, `input_package/scenarios/severe.csv`
- **Outputs:** `outputs/support/documentation_crosscheck.md`, `outputs/support/findings_register.json`
- **Result:** Found severe house-price growth less severe than adverse in 4 quarter(s): 2026Q3, 2026Q4, 2027Q1, 2027Q2.

EVT-05 - Reconcile horizon and reversion descriptions

- **Phase:** documentation_challenge
- **Related procedure:** GA-04
- **Action:** Compared methodology, model overview, and scenario package to determine whether one consistent forecast and reversion structure was evidenced.
- **Inputs:** `input_package/docs/methodology.md`, `input_package/docs/model_overview.md`, `input_package/docs/scenario_assumptions.md`

- **Outputs:** `outputs/support/documentation_crosscheck.md`, `outputs/support/findings_register.json`
- **Result:** Identified inconsistent 8Q/4Q and 6Q/2Q horizon descriptions with only six numeric quarters provided.

EVT-06 - Reconcile documented segments to output segments

- **Phase:** documentation_challenge
- **Related procedure:** GA-05
- **Action:** Compared documented segment taxonomy to the segments used in the provided reserve output files.
- **Inputs:** `input_package/docs/methodology.md`, `input_package/docs/model_overview.md`, `input_package/outputs/provided_segment_reserves.csv`
- **Outputs:** `outputs/support/findings_register.json`
- **Result:** Documentation lists 5 segments while provided outputs reconcile to 4 segments.

EVT-07 - Reconcile documented overlay cap to provided overlay bridge

- **Phase:** documentation_challenge
- **Related procedure:** GA-06
- **Action:** Compared the documented overlay cap and governance description to the segment overlay values in the provided bridge.
- **Inputs:** `input_package/docs/overlay_memo.md`, `input_package/outputs/provided_overlay_bridge.csv`
- **Outputs:** `outputs/support/findings_register.json`
- **Result:** Documented overlay cap of 6.0 bps did not reconcile to provided overlay values reaching 18.0 bps.

EVT-08 - Record blocked baseline reproduction

- **Phase:** blocked_work
- **Related procedure:** GA-07
- **Action:** Evaluated whether baseline reproduction could be performed and formally logged why it was not executed.
- **Inputs:** `input_package/outputs/provided_reserve_summary.csv`, `input_package/docs/evidence_reque`
- **Outputs:** `outputs/support/coverage_statement.md`, `outputs/support/evidence_request_list.md`
- **Result:** Baseline reproduction blocked because no reserve engine or reproducibility notebook was supplied.

EVT-09 - Record blocked scenario reruns and sensitivity testing

- **Phase:** blocked_work
- **Related procedure:** GA-08
- **Action:** Evaluated whether scenario reruns and sensitivity tests could be performed and formally logged why they were not executed.
- **Inputs:** `input_package/scenarios/*.csv`, `input_package/docs/gap_tracker.md`
- **Outputs:** `outputs/support/coverage_statement.md`, `outputs/support/evidence_request_list.md`
- **Result:** Sensitivity testing blocked because no model implementation, controllable parameters, or runbook support was supplied.

EVT-10 - Assemble internal gap-assessment pack

- **Phase:** synthesis
- **Related procedure:** GA-09
- **Action:** Combined discovery, coverage boundary, documentation findings, and evidence requests into the final internal gap-assessment report pack.
- **Inputs:** `outputs/support/review_strategy.md`, `outputs/support/procedure_run_log.md`, `outputs/support/findings_register.json`, `outputs/support/evidence_request_list.md`
- **Outputs:** `outputs/stakeholder/cecl_gap_assessment.tex`, `outputs/stakeholder/cecl_gap_assessment`
- **Result:** Produced a comprehensive CECL gap assessment showing both executed documentation-led work and blocked runtime work.

Provided Output Snapshot

Although no executable reserve engine was supplied, the package included prior reserve summaries and segment output snapshots that could be reconciled at a high level.

Scenario	Provided Reserve	Provided Reserve Rate
Baseline	$38,420,000	1.84%
Adverse	$52,180,000	2.48%
Severe	$53,010,000	2.51%

Output Segment	Baseline	Adverse	Severe
Commercial And Industrial	$12,920,000	$17,310,000	$17,720,000
Cre Investor	$11,900,000	$17,120,000	$17,880,000
Heloc	$5,180,000	$6,450,000	$6,430,000
Residential Mortgage	$8,420,000	$11,300,000	$10,980,000

Segment	Provided Overlay
Cre Investor	18.0 bps
Heloc	16.0 bps
Commercial And Industrial	14.0 bps
Residential Mortgage	12.0 bps

Documentation and Evidence Assessment

Execution Readiness

The package does not support runtime validation. No reserve engine, no reproducibility notebook, and no lineaged runbook were included, so the review could not test whether the provided reserves are reproducible from the documented methodology.

Scenario Consistency

The numeric scenario files cover six quarters, not the full documented forecast and reversion path. In addition, the severe scenario is not uniformly harsher than the adverse scenario for house-price growth in the following quarters: 2026Q3, 2026Q4, 2027Q1, 2027Q2.

Segment and Overlay Reconciliation

Documentation references 5 segments, while the provided reserve output files reconcile to 4 segments. The overlay memo frames qualitative adjustments as capped at 6.0 bps, but the supplied bridge reaches 18.0 bps.

Detailed Gaps and Findings

Execution-based review is blocked by missing reserve engine and lineage evidence

Severity: HIGH
Observation: The package includes prior outputs and narrative materials, but no reserve engine, reproducibility script, or lineaged runbook sufficient for execution-based review.
Evidence: docs/evidence_request_log.md, docs/gap_tracker.md, outputs/provided_reserve_summary.csv
Recommended response: Provide executable model artifacts, a reproducibility notebook or script, and a lineaged runbook before requesting an execution-based CECL opinion.

Scenario narrative is not fully aligned to the supplied scenario tables

Severity: HIGH
Observation: The severe scenario narrative describes a uniformly harsher path, but the supplied numeric scenarios show 4 quarter(s) where severe house-price growth is less severe than adverse.
Evidence: docs/scenario_assumptions.md, scenarios/adverse.csv, scenarios/severe.csv

Recommended response: Refresh the scenario narrative so it matches the numeric path quarter by quarter and include governance approval for the final scenario set.

Documented segment structure does not reconcile to the provided reserve outputs

Severity: MEDIUM
Observation: Documentation describes 5 segments, while the provided reserve outputs reconcile to 4 output segments.
Evidence: docs/methodology.md, docs/model_overview.md, outputs/provided_segment_reserves.csv
Recommended response: Reconcile segment taxonomy across methodology, model overview, and reserve reporting, then restate the official segment inventory in one controlled artifact.

Overlay documentation understates the magnitude implied by the provided reserve bridge

Severity: MEDIUM
Observation: The overlay memo frames qualitative adjustment as capped at 6.0 bps, but the provided bridge shows up to 18.0 bps.
Evidence: docs/overlay_memo.md, outputs/provided_overlay_bridge.csv
Recommended response: Quantify and document the overlay methodology and cap so the reserve bridge can be independently reconciled.

Evidence Requests and Next Actions

The following items are required before the package can support execution-based review procedures.

1. Provide the runnable CECL reserve engine or reproducibility notebook used to generate the supplied reserve outputs.
2. Provide a lineaged runbook linking the prior output snapshots to the model version and scenario definitions reviewed by governance.
3. Reconcile documented segment taxonomy to the segment structure used in the reserve outputs.
4. Refresh scenario documentation so the severe narrative matches the numeric severe path quarter by quarter.

Overall Assessment

The package is sufficient for a structured CECL gap assessment, but not for a model-driven review opinion. The principal blockers are missing execution artifacts, inconsistent horizon and segment documentation, scenario narratives that do not fully align to numeric tables, and overlay support that is not quantitatively reconciled.